Traffic STGNNs across Sensor, City, and Time Shifts: Routing Concentration Tracks Sensitivity to Sensor Dropout

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Abstract

Traffic forecasting benchmarks leave open how architectures behave when sensors fail, models move between cities, or weekday and weekend traffic differ. We evaluate eight STGNN architectures under these three conditions using frozen checkpoints and matched access to source-city topology. Sensitivity to silent zero-filling is associated with learned routing concentration. Across six adaptive backbones trained on METR-LA, including GTS and DGCRN added for this analysis, the correlation between mean row-maximum routing mass and dropout degradation is Pearson $\rho{=}+0.974$; it is positive but weaker on PEMS-BAY and PEMS04. Graph WaveNet degrades sharply, while AGCRN and STAEformer stay close to DCRNN. When failed sensors are known, carrying forward the last observed value (LOCF) removes most of the gap. Rankings differ under the other conditions: MegaCRN leads source-normalized cross-city transfer, while MegaCRN and STAEformer have positive normalized weekday/weekend gaps but lower weekend error in masked mph. We release a frozen-checkpoint registry, evaluation code, and a decision table summarizing model failure modes.

1 Introduction

Traffic forecasting supports urban services such as ride-hailing dispatch, signal timing, and congestion management. STGNNs are now standard tools [Li et al., 2018, Wu et al., 2019, Yu et al., 2018], but they are usually selected with training and test data from the same city, sensors, and time window. In deployment, sensors can fail silently, a model trained in one city may be applied to another, and traffic changes between weekdays and weekends. Clean MAE alone does not indicate how well a model handles these changes.

Prior STGNN robustness work proposes OOD interventions via retrieval or pruning [Zhang et al., 2025, Jing et al., 2026] or frames cross-city transfer as federated learning [Zeng et al., 2025]. A controlled *diagnostic* is still needed: the same frozen backbone evaluated under several deployment shifts, with differences traced to architectural choices. We also pre-register hypotheses so failed predictions remain part of the evidence (Appendix A). Cross-city comparisons need one more control: adaptive models can in principle re-learn a target topology, while fixed-graph models cannot. A matched-structural-access protocol, where every backbone uses source-city topology, removes that asymmetry. We use this protocol to study when model rankings change across shifts and to examine the architectural mechanisms behind those changes.

We assemble a bank of eight architectures trained on METR-LA: a five-model canonical bank spanning graph-free (GRU), fixed-graph (DCRNN, STGCN), and adaptive-graph (MTGNNLight, GWNet) models, plus three modern adaptive-relation models (AGCRN, MegaCRN, STAEformer).

40th Conference on Neural Information Processing Systems (NeurIPS 2026). Track on Evaluations and Datasets.

Each frozen checkpoint is evaluated under three shifts: **sensor dropout** (10–50% zeroed at inference, 5 random masks), **cross-city transfer** (zero-shot METR-LA $\rightarrow$ PEMS-BAY under matched structural access), and **temporal subgroup** (weekend vs. weekday windows within the METR-LA test set). The sensor experiment targets silent zero-filling; when failed sensors are known, carrying forward the last observed value (LOCF) removes most of the gap (Appendix C.8). We report temporal subgroup differences in both normalized and physical units, since the two conventions can reverse the apparent direction of a model's gap.

Our main contributions are as follows:

- We introduce a frozen-checkpoint evaluation protocol for traffic STGNNs under sensor, city, and temporal shifts, with matched structural access for cross-city comparison.
- We show that routing concentration tracks sensor-dropout sensitivity across adaptive backbones and test this link on Graph WaveNet with inference-time interventions.
- We provide a shift-specific robustness taxonomy and a recommendation artifact to support model selection across deployment conditions and evaluation conventions.

Code and artifacts are available at `https://github.com/sirjuststop/stgnn-shift-bench`.

2 Related Work

Evaluation and robustness benchmarks. NeurIPS Datasets and Benchmarks papers such as BEIR [Thakur et al., 2021] and RobustBench [Croce et al., 2021] extend model evaluation to generalization and robustness. The closest spatiotemporal analogue is Yi et al. [2021]'s corruption benchmark for video models. Here, we focus on traffic forecasting: the same frozen STGNNs are audited under sensor failure, city transfer, and temporal subgroup shift. Traffic OOD work mostly proposes adaptation or retrieval methods, and spatiotemporal benchmarks test corruption robustness or zero-shot transfer; neither pairs frozen-architecture diagnostics with mechanism analysis.

Spatiotemporal graph neural networks for traffic. DCRNN [Li et al., 2018], STGCN [Yu et al., 2018], Graph WaveNet [Wu et al., 2019], GRU, and MTGNNLight span graph-free, fixed-graph, and adaptive-graph designs in our canonical bank. We extend the diagnostic to AGCRN [Bai et al., 2020], MegaCRN [Jiang et al., 2023], and STAEformer [Liu et al., 2023a], all competitive traffic forecasters with learned spatial or temporal relation modules. The comparison separates topology learning from routing concentration: MTGNNLight's sparse learned graph is less fragile than GWNet's dense softmax graph, while AGCRN and STAEformer match DCRNN under sensor dropout.

Distribution shift in spatiotemporal models. Recent methods address distribution shift in STGNNs. STRAP [Zhang et al., 2025] and TL-GPSTGN [Jing et al., 2026] propose OOD interventions through retrieval or graph-context pruning. We examine how sensitivity varies across backbones and shifts, and which architectural choices account for those differences.

Cross-city transfer learning. Cross-city work often assumes federated or retrieval-based adaptation, as in FedTT [Zeng et al., 2025] and CAST-CKT [Fofanah et al., 2026]; large-scale benchmarks such as LargeST [Liu et al., 2023b] address scale rather than zero-shot transfer. We evaluate the stricter zero-shot case: models trained only on METR-LA applied directly to PEMS-BAY, with no target-city adaptation.

Topology-agnostic and foundation-model approaches. Topology-free time-series foundation models such as Chronos [Ansari et al., 2024] adapt through pretraining or fine-tuning rather than explicit road topology. Our zero-shot setting asks which architectural properties reduce fragility before any adaptation is available.

Sensor failure and data imputation. Mask-aware reconstruction architectures such as GRIN [Cini et al., 2022] and BRITS [Cao et al., 2018] reconstruct missing inputs before or during graph aggregation. We compare how standard STGNN architectures respond to missing inputs without imputation. LOCF nearly eliminates the error differences (Appendix C.8), so the sensor result is specific to silent zero-masking.

3 Problem Setup and Backbone Bank

The audit evaluates eight frozen STGNN architectures under three deployment-motivated shifts. Table 1 summarizes each shift axis, its metric, the protocol control, the main validity threat, and the appendix that addresses it.

Table 1: **Audit design.** Three deployment-motivated conditions evaluated with the same frozen checkpoints.

Shift axis	Metric	Control	Validity threat	App. check
Sensor dropout	Δ_{sensor}: MAE delta (norm.)	MAE on kept sensors only; primary mask	Random vs. structured dropout pattern	C.4, C.3
Cross-city transfer	X-city norm. MAE	Matched structural access (source adj)	Subset sampling; padding inflation	D.2, D.6
Temporal subgroup	Δ_{temp}: weekend−weekday (norm.; mph comparison)	Within-split stratification; no new data	Metric-dependent sign reversal	E.1, E.3

How to read the audit. Each column measures a separate quantity: clean MAE identifies the checkpoint, Δ_{sensor} measures damage from silent zero-masking, cross-city MAE measures zero-shot transfer under matched structural access, and Δ_{temp} measures a within-split weekday/weekend gap. We compare models within each axis; the columns do not form a single leaderboard score.

3.1 Backbone Architectures

The canonical bank covers five architectures from graph-free to fixed-graph to adaptive-graph.

DCRNN [Li et al., 2018] uses a bidirectional diffusion GRU on a fixed road-distance adjacency $\mathbf{A}$ (truncated at K hops in both directions), fixed at training and inference. **Graph WaveNet (GWNet)** [Wu et al., 2019] adds a learned adaptive adjacency

$$\hat{\mathbf{A}} = \text{softmax}(\mathbf{E}_1 \mathbf{E}_2^\top), \tag{1}$$

with $\mathbf{E}_1, \mathbf{E}_2 \in \mathbb{R}^{N \times d_e}$ trained end-to-end and tied to the N training nodes. **STGCN** [Yu et al., 2018] alternates gated temporal convolutions with truncated-Chebyshev spectral graph convolutions over the same fixed road adjacency. **GRU** is a graph-free recurrent baseline that treats the N sensors as a single feature vector and uses no adjacency. **MTGNNLight** is a lightweight MTGNN variant [Wu et al., 2020] that learns a sparse adaptive graph via pairwise cosine similarity on a single embedding $\mathbf{E} \in \mathbb{R}^{N \times d}$ and keeps only top-k neighbors per node. This directly contrasts with GWNet's dense all-pairs softmax and isolates whether dropout fragility depends on adjacency density or on topology learning per se (P14).

These five share a unified config (hidden dim 64, seq/pred 12, Adam 1×10^{-3}, weight decay 10^{-4}, cosine schedule, 100 epochs, patience 15). To rule out training-artifact explanations of GWNet's fragility, we additionally train **GWNet[CM]** (patience 30) and **GWNet[Strong]** (P17; hidden 128, adaptive embed 20, patience 30); both are frozen at shift-eval time.

3.2 Backbone Bank Protocol

We train each backbone once on METR-LA [Li et al., 2018] using the chronological 70/10/20 split and three fixed seeds. The resulting checkpoints are frozen before any shift evaluation: no model is re-trained, fine-tuned, or selected separately for sensor dropout, cross-city transfer, or temporal subgroup analysis. An additional four-backbone few-shot probe relaxes this constraint to test limited target-data adaptation (Appendix D.4).

Table 2 reports the clean METR-LA MAE together with the three shift metrics. The clean column is included only to anchor the starting point of each checkpoint. Our claims are not based on clean ranking: a backbone can be useful for diagnosing a failure mechanism even when it is not the

strongest source-city forecaster. For the same reason, GWNet[CM] and GWNet[Strong] are treated as control checkpoints for the GWNet finding, not as additional entries in the main bank.

3.3 Extended adaptive-relation models

To probe generality, we add three architectures using the same METR-LA split and frozen-checkpoint rule: **AGCRN** [Bai et al., 2020] (node-adaptive GRU, no temporal covariates); **MegaCRN** [Jiang et al., 2023] (meta-graph memory with TOD/DOW inputs); **STAEformer** [Liu et al., 2023a] (Transformer with spatial-temporal adaptive embeddings encoding temporal covariates). These models use their standard masked-MAE convention, while the canonical bank uses plain MAE; Table 3 reports clean values only to identify checkpoints, not to rank clean performance across families.

We also include GTS [Shang et al., 2021] and DGCRN [Li et al., 2023] for routing concentration and sensor severity, outside the eight-model cross-city/temporal bank. Native-source routing analyses also cover PEMS-BAY and PEMS04.

3.4 Distribution shifts

Sensor dropout. At inference, a random 30% of sensor channels are zero-masked. We do not re-instantiate the 207-node model; it sees a 207-dimensional input with zeroed channels, and MAE is computed on the kept sensors only so clean and masked evaluations are directly comparable. Tables 2–3 report the pre-registered primary-mask aggregate; Appendix C.3 reports 5-mask aggregates to separate mask-sampling from model-weight variance. GRU is the main mask-sampling caveat. Structured dropout patterns (hub and contiguous) are in Appendix C.4. This protocol isolates silent failure; the LOCF comparison in Appendix C.8 shows that effective imputation can largely remove the zero-mask gap when the true failure mask is available. Supplementary probes add stale hold, noise, bias, and intermittent missingness (Appendix C.2).

Cross-city transfer. The frozen METR-LA model is evaluated zero-shot on PEMS-BAY [Li et al., 2018] (Bay Area, 325 sensors, 52,116 timesteps). A second pair (PEMS04 source, 307 sensors; zero-shot on METR-LA and PEMS-BAY) is in Appendix D.5 for cross-city generality (P18). "Zero-shot" means no re-training or fine-tuning. Because PEMS-BAY has 325 sensors and the METR-LA checkpoints have 207 input channels, we subsample 207 PEMS-BAY sensors (5 subsets); rankings remain stable at 150-node coverage (Appendix D.2).

Matched structural access (canonical comparison). The main cross-city comparison uses source-only adjacency: every backbone uses the METR-LA adjacency at inference, removing the asymmetry that adaptive-graph models cannot access target topology by construction. Under this constraint, differences between backbones cannot come from privileged access to target topology (target-adj results: Appendix D.3). MAE is normalized with source-city statistics for cross-architecture comparability.

Temporal subgroup shift. METR-LA spans weekdays and weekends with different traffic patterns; the model is trained on both. We stratify the test set by day type—weekday (5,104 windows) and weekend (1,728 windows)—and report $\Delta_{\text{temp}} = \text{MAE}_{\text{weekend}} - \text{MAE}_{\text{weekday}}$. This is a subgroup performance gap within the existing test split, not deployment to a new time window. In addition to the normalized gaps, we report the extended models' masked-mph gaps on the same checkpoints and windows (Appendix E.1). The latter exclude zero-valued targets, whereas normalized MAE includes all targets; the comparison changes both scaling and the target mask.

Aggregation. Main tables report mean $\pm$ population std over three training seeds; additional mask and subset averages are labeled separately. The canonical bank uses METR-LA train-statistics normalization, and extended models use their model-specific scalers (§3.3).

4 Robustness Taxonomy

Pre-registered hypotheses. We pre-registered 16 predictions in five stages before running the corresponding experiments. Of these, 10 were confirmed, 4 were refuted, and 2 were partially confirmed. The full prediction–outcome ledger is in Appendix A; the relevant confirmed predictions

(P14–P19, P21) are referenced inline where their results appear. We use the ledger to keep the taxonomy falsifiable: failed predictions remain in the appendix, and main-text claims are tied either to a pre-specified contrast or to a control experiment that removes a concrete artifact.

Table 2 reports the main results across all five backbones and three shift types. The columns should not be averaged into a single robustness score. They use different estimands and normalizers, so the evidence is the change in ordering from one deployment axis to the next.

Table 2: **Robustness rankings are shift-specific.** Five backbones evaluated across sensor dropout, cross-city transfer, and temporal subgroup shift. Values are mean ± population std over 3 seeds; lower is better.

Backbone	In-dist MAE (mph)	Δ_{sensor} (30%, norm.)	X-city (norm.)†	Δ_{temp} (norm.)
DCRNN (fixed diffusion)	6.58 ± 0.03	$+0.020 \pm 0.002$	0.180 ± 0.001	$+0.087 \pm 0.004$
GRU (graph-free)	7.49 ± 0.12	$+0.025 \pm 0.012$	0.278 ± 0.009	$+0.093 \pm 0.003$
MTGNNLight (sparse cosine-sim)	6.88 ± 0.11	$+0.051 \pm 0.012$	0.244 ± 0.004	$+0.075 \pm 0.007$
STGCN (fixed spectral)	7.12 ± 0.05	$+0.078 \pm 0.007$	0.249 ± 0.008	$+0.093 \pm 0.003$
GWNet (dense adaptive adj)	7.85 ± 0.11	$+0.088 \pm 0.017$	0.268 ± 0.006	$+0.094 \pm 0.009$

†Source-only adjacency; METR-LA-stat normalized MAE on PEMS-BAY.

Sensor dropout. The sensor axis separates GWNet from the other canonical backbones. DCRNN and GRU have the smallest deltas, while MTGNNLight is markedly less fragile than GWNet despite also learning topology. Learning a graph is therefore not enough to explain the fragility; the relevant contrast is sparse learned routing versus dense all-pairs softmax. Capacity controls reduce but do not remove the GWNet gap (Table 7; Appendix G.1). The same pattern holds under hub dropout, which widens GWNet's gap while making DCRNN less sensitive, which is the expected direction if dropped high-degree nodes expose concentrated learned routes rather than merely increasing the number of missing inputs. The ranking also depends on corruption type: at 30% affected channels, GWNet's healthy-sensor increase is $+0.16$ mph for noise but $+1.96$ mph for bias, compared with DCRNN's $+0.28$ and $+1.00$ mph, respectively (Appendix C.2).

Cross-city transfer under matched structural access. Under source-only adjacency, DCRNN is the strongest canonical-bank transfer baseline. The result is not a simple fixed-vs-adaptive split: STGCN loses its target-adjacency advantage once target topology is withheld, and then performs similarly to GWNet under the fair comparison. The DCRNN advantage also holds on a second city pair, while padding sensitivity changes absolute MAE but not the ordering (Table 7; Appendix D.6). The GRU baseline is useful here because it removes graph mismatch entirely and still transfers worst in the canonical bank. Cross-city robustness therefore does not come from ignoring structure; it depends on how the source-city structure is used when the target graph is unavailable. A limited-adaptation probe is reported in Appendix D.4.

Temporal subgroup shift (canonical bank). Within the canonical bank, weekend/weekday gaps are effectively tied: the between-model spread is small relative to the sensor and city-transfer separations, and it is not aligned with graph type. None of these architectures condition on explicit time-of-day or day-of-week, and the gaps show no dependence on graph type. We therefore treat the canonical-bank temporal column as a negative result and as the baseline for §4.1, where reporting conventions change the interpretation of the extended models' temporal gaps.

Shift-specificity (canonical bank). DCRNN leads source-only city transfer, while the sensor-dropout and temporal rankings differ from it; the adaptive-relation models in §4.1 change the transfer ordering again. Capacity, source-adjacency, structured dropout, and horizon controls are consolidated in Appendix Table 7.

4.1 External validity: modern adaptive-relation models

The canonical bank covers four spatial-aggregation strategies but omits the newer adaptive-relation designs used in recent traffic forecasting work. To test whether the findings carry over, we add three architectures (AGCRN, MegaCRN, STAEformer) trained on the same METR-LA data with the same split, frozen after training, evaluated under the same protocol. Table 3 reports results alongside DCRNN as a reference.

Table 3: **Adaptive-relation models under the three shift axes.** Clean MAE follows each architecture's standard masked-MAE convention; shift columns use the same normalized protocol as Table 2.

Model	Clean MAE (mph)	Δ_{sensor} (30%, norm.)	Cross-city (src, norm.)	Δ_{temp} (norm.)
AGCRN	$4.09 \pm 0.02^{\dagger}$	$+0.019 \pm 0.001$	0.206 ± 0.009	$+0.061 \pm 0.008$
MegaCRN	$3.04 \pm 0.06^{\dagger}$	-0.048 ± 0.027	0.163 ± 0.003	$+0.734 \pm 0.019$
STAEformer	$2.93 \pm 0.01^{\dagger}$	$+0.015 \pm 0.001$	0.223 ± 0.008	$+0.750 \pm 0.003$
DCRNN (ref)	6.58 ± 0.03	$+0.020 \pm 0.002$	0.180 ± 0.001	$+0.087 \pm 0.004$

† Masked MAE (target > 0); DCRNN uses plain MAE. Cross-family clean-performance ranking is not meaningful due to this convention difference.

Sensor dropout. AGCRN and STAEformer stay near DCRNN on sensor dropout, so GWNet's result cannot be read as "adaptive graphs are fragile." The distinction is dense all-pairs softmax (Eq. 1) versus node-specific or attention-mediated relations. MegaCRN's negative delta is non-monotonic in dropout ratio and is treated conservatively as adjacency-mediated smoothing toward a weekday prior (§5.3), not as direct dropout robustness.

Cross-city transfer. MegaCRN attains the lowest source-stat cross-city MAE, so DCRNN's lead in zero-shot transfer holds only within the canonical bank. AGCRN and STAEformer fall between DCRNN and GWNet.

Temporal subgroup shift. In physical units, MegaCRN and STAEformer have *lower* weekend error: their weekend-minus-weekday gaps are -0.594 ± 0.051 and -0.612 ± 0.002 masked mph, respectively. AGCRN's gap is $+0.750 \pm 0.013$ mph. The normalized gaps in Table 3 are positive for all three models. Thus the large normalized gaps do not establish greater physical-error fragility: scaling and target masking reverse the sign for two models (Appendix E.1). Covariate interventions address the normalized gap, not this absolute-error ranking.

Comparison across shifts. The adaptive-relation models differ across deployment settings (Figure 1): AGCRN and STAEformer match DCRNN under sensor dropout, MegaCRN leads source-normalized cross-city transfer, and temporal comparisons depend on the reporting convention. Routing concentration tracks sensor-dropout sensitivity even when clean accuracy and cross-city ranking move in different directions. The sensor-dropout ranking is specific to silent zero-masking; under last-observed-value carry-forward (LOCF) with the true failure mask, the four backbones in the comparison recover $\geq 97\%$ of the gap (Appendix C.8), so the practical impact of GWNet's fragility depends on the deployment pipeline's imputation policy. The main conclusions are drawn from METR-LA as the source city; PEMS04 is used only for targeted source-city replication, not as a full third-source benchmark (§6).

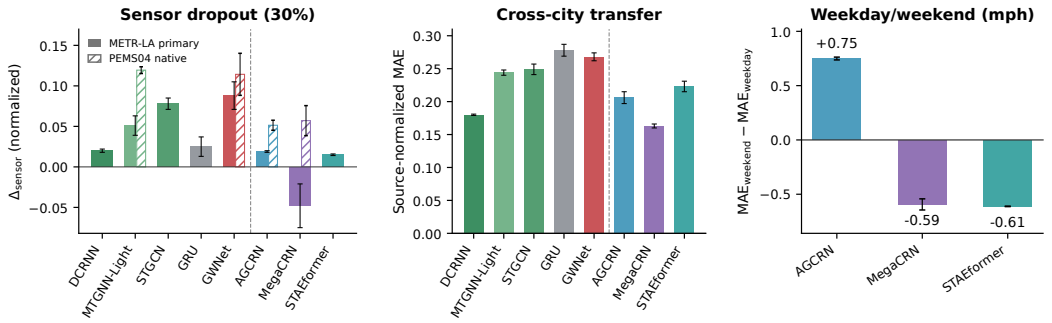


Figure 1: **Shift profiles and the absolute-unit temporal correction.** Sensor and city panels show all eight backbones; the dashed line separates the canonical and extended banks. Hatched bars show PEMS04-native replication. The temporal panel shows masked-mph gaps for the three extended models: MegaCRN and STAEformer have lower weekend error despite positive normalized gaps. Error bars are population std over three seed-level means.

5 Mechanism: dense adaptive routing and sensor-dropout fragility

The taxonomy shows that GWNet's dense learned routing is fragile under sensor dropout, whereas AGCRN and STAEformer, which also learn spatial relations, are not (Table 3). We analyze GWNet because it exhibits the failure; an ablation on a robust backbone would leave little fragility to attribute. We test whether the dense adaptive adjacency (Eq. 1) drives the failure and whether it can be reduced without retraining.

5.1 Ablation: adaptive vs. fixed adjacency

We run frozen GWNet checkpoints under two inference-time adjacency configurations. **GWNet[adaptive]** keeps $\hat{\mathbf{A}} = \text{softmax}(\mathbf{E}_1\mathbf{E}_2^\top)$; **GWNet[fixed]** replaces $\hat{\mathbf{A}}$ with the normalized METR-LA distance adjacency $\mathbf{A}$ and zeros out $\mathbf{E}_1, \mathbf{E}_2$ (rest of the network unchanged).

Table 4: **Fixed-road GWNet reduces sensor-dropout sensitivity and city-transfer MAE.** Mean $\pm$ population std over 3 seeds; city MAE is target-adj PEMS-BAY mph.

Variant	In-dist MAE	Δ_{sensor} (30%, primary mask)	City MAE (mph)
GWNet[adaptive]	7.85 ± 0.11	$+0.088 \pm 0.017$	4.42 ± 0.15
GWNet[fixed]	8.27 ± 0.08	$+0.050 \pm 0.010$	3.44 ± 0.04
Change	$+5.3\%$	-43.1%	-22%

Fixed substitution cuts sensor-dropout sensitivity by 43.1% and city-transfer MAE by 22%, at $+5.3\%$ in-distribution cost, past the pre-registered $\geq 30\%$ threshold for attributing fragility to the adaptive adjacency. Per-horizon decomposition (Table 7) shows GWNet's cross-city MAE is nearly horizon-flat ($+4\%$ growth h=1→12) while DCRNN grows $+46\%$, consistent with a static routing mismatch rather than error that accumulates over the horizon.

Mask-aware adjacency variant. Zeroing and renormalizing adjacency rows/columns for dropped sensors at inference ($\hat{\mathbf{A}}[i,:]$=$\hat{\mathbf{A}}[:,i]$=0, then $\hat{\mathbf{A}} \leftarrow \hat{\mathbf{A}}/\max(\hat{\mathbf{A}}\mathbf{1}, \epsilon)$) should partially restore robustness without retraining.

Table 5: **Mask-aware routing reduces GWNet's multi-mask dropout gap.** Dropped-sensor rows/columns are zeroed and renormalized; each of 3 seeds is averaged over 5 random 30% masks. Error bars summarize the seed means; recovery is the mean of per-seed recovery ratios.

Variant	Δ_{sensor} (30%, multi-mask)	Recovery
GWNet[standard]	$+0.092 \pm 0.030$	—
GWNet[mask-aware]	$+0.023 \pm 0.009$	75.2%

Mask-aware routing recovers 75.2% of the multi-mask gap, whereas fixed substitution reduces the primary-mask gap by 43.1%. A plausible reason is that masking only the affected rows and columns keeps the remaining learned routes, which fixed substitution discards. These percentages use different mask protocols. The intervention is $\sim$25 LoC and can be applied to existing GWNet checkpoints. The recovery is stable across dropout ratios: 70–76% for random 10–40% dropout and 71% for hub-30% dropout (Appendix C.7).

5.2 Routing concentration as a shared mechanism hypothesis

The GWNet analysis above provides model-specific evidence; we next examine the association over the four adaptive-graph models in our bank (n=4 adaptive backbones). We measure adjacency concentration as max_mass, the mean per-row maximum routing probability, using each model's native relation proxy. Pairing this proxy with the routing-spectrum Δ_{sensor} estimand gives Pearson ρ=+0.979 (Appendix F.2). Removing GWNet leaves ρ=+0.961; the exact permutation test gives p=0.083 (n=4; Appendix F.2.2). Seed-level and rank diagnostics are in Appendix F.2.1. STAEformer

has no single row-stochastic adjacency and is excluded from this proxy comparison. The main tables retain their primary-mask estimand; substituting those deltas reduces the four-model correlation to $+0.880$. We extend the architecture sample with GTS [Shang et al., 2021] (link-predictor graph learning) and DGCRN [Li et al., 2023] (input-conditioned dynamic graphs), selected for their distinct mechanisms. Across six architectures, ρ=$+0.974$; dropping GWNet leaves $+0.973$, and the minimum leave-one-out value is $+0.961$. Native-source associations are positive but weaker on PEMS-BAY ($+0.75$) and PEMS04 ($+0.63$; Appendix F.3). The original four-model relation also replicates on native PEMS04 checkpoints (ρ=$+0.910$; Appendix F.4). Together, these comparisons support a shared-mechanism hypothesis within the evaluated architectures and traffic datasets. A local perturbation argument supports this: zero-masking a dropped sensor j shifts each kept node i by $A_{ij}H_jW$; averaging over rows bounds the mean signal disruption by $B_H B_W |\mathcal{I}| \max_\text{mass}(A)$. Concentrated routing raises this loose bound proportionally; we treat $\max_\text{mass}$ as a tail-risk proxy for iid dropout. This is a single-layer perturbation bound, not a full-horizon MAE theorem. The complete derivation and caveats are in Appendix F.5.

Within GWNet, top-k sparsification at inference traces the same curve: relative fragility rises from 0.80% at k=10 to 1.17% at k=full (Appendix F.1).

5.3 Ruling out two natural mechanism hypotheses

Two ablations probe the normalized MegaCRN/STAEformer temporal gaps. Randomizing or swapping explicit time embeddings leaves MegaCRN's normalized gap unchanged and reduces STAEformer's by only 12.5%, so TOD/DOW covariates alone do not explain that metric. MegaCRN's negative sensor delta is adjacency-mediated smoothing toward a weekday-dominated prior, not evidence that MegaCRN directly tolerates dropout (Appendix G.2). These interventions do not establish excess temporal error in mph, where the gaps reverse sign. Isolating the remaining normalized-gap mechanism requires retraining, left to future work (Appendix E.3).

5.4 Severity, degree stratification, and imputation baselines

In the original three-model severity comparison, GWNet is the most sensitive backbone at all five dropout ratios (GWNet/DCRNN ratio grows $2.5\times \rightarrow 4.6\times$ from 10% to 40%; Appendix C.1). The extended physical-unit comparison places DGCRN, the highest-concentration model, at the most fragile endpoint at four of five severities (Table 10). GWNet's highest-degree quartile suffers Δ_{sensor}=$+0.121$ versus $+0.066$ for Q1 ($1.83\times$); DCRNN shows no degree dependence (Appendix C.5). Last-observed-value carry-forward (LOCF) recovers $\geq 97\%$ of the zero-mask gap across the four backbones in that comparison, so the ranking is specific to silent-failure zero-masking (Appendix C.8).

6 Discussion

Deployment guidance. Within the shifts, source city, and masking model evaluated here, the results suggest several practical choices. Under silent-failure zero-mask dropout, DCRNN, AGCRN, and STAEformer degrade much less than GWNet; the GWNet gap is tied to dense softmax routing and the mask-aware variant recovers 75% without retraining. For zero-shot cross-city transfer on METR-LA/PEMS-BAY-style highway networks, MegaCRN and DCRNN perform well. Weekday/weekend comparisons require an explicit metric: MegaCRN and STAEformer have positive normalized gaps but negative masked-mph gaps. Their normalized values alone should not drive temporal deployment guidance.

Implications for robustness evaluation. In our bank, the clean-test ranking changes under sensor dropout and only partly predicts cross-city transfer. Richer relation learning is not uniformly more robust; weaknesses appear on different axes. Benchmarks should therefore report per-shift results alongside in-distribution accuracy. The GWNet controls make simple optimizer failure an unlikely explanation for its fragility (Table 7).

Recommendation artifact. The released $2\times2\times2$ decision table maps sensor reliability, city transfer, and weekend exposure to observed fragility and mitigations (Appendix H). It covers the eight-model bank and demotes MegaCRN's smoothing-related negative sensor delta. Its temporal axis uses

normalized gaps. AGCRN, recommended for two weekend-exposed profiles, has the only positive masked-mph weekend gap among the extended models (Table 24); these cells should not be used to choose models for physical-error performance.

Limitations. All datasets are highway loop-detector data. METR-LA is the main training source, with PEMS-BAY and PEMS04 used for targeted replications; urban arterials and probe data remain untested. The model set is not exhaustive (e.g., ASTGCN [Guo et al., 2019], DSTAGNN [Lan et al., 2022]), and the main results use three training seeds. Zero-shot transfer is a diagnostic floor, not a deployment recommendation; the few-shot probe covers four backbones and limited adaptation (Appendix D.4). The weekday/weekend comparison is a subgroup analysis within one test split, not an OOD test. Walk-forward evaluation remains future work, and the gap's sign depends on normalization and target masking. The six-architecture concentration correlation weakens on PEMS-BAY and PEMS04; interventions support the link only for GWNet. MegaCRN's negative sensor delta reflects adjacency-mediated smoothing, and TOD/DOW covariates alone do not explain its normalized temporal gap.

Scope of the zero-mask diagnostic. Zero is a physically possible speed, so a zero reading alone cannot identify failure. Corruptions are assigned by a failure mask, not by observed values; valid zeros at kept sensors are not classified as failures. Both remedies require identifying failed sensors: LOCF uses the true failure mask (Appendix C.8), and detector errors increase MAE for mask-aware routing (Appendix C.6). We did not test LOCF under imperfect detection. The gap applies where failures go undetected, telemetry is delayed, or carry-forward is unreliable.

7 Conclusion

Across six adaptive architectures trained on METR-LA, sensitivity to silent zero-filling correlates with routing concentration (Pearson $\rho{=}+0.974$); the association is positive but weaker on PEMS-BAY and PEMS04. For Graph WaveNet, mask-aware inference recovers 75% of the dropout gap without retraining when failed sensors are known. Cross-city and temporal evaluation give different rankings: MegaCRN leads source-normalized transfer, and the sign of MegaCRN's and STAEformer's weekday/weekend gap depends on whether errors use normalized units or masked mph. Robustness claims for traffic STGNNs should name both the shift and the metric.

Acknowledgments and Disclosure of Funding

This work was supported by the National Natural Science Foundation of China (Grant No. 62506134), the Guangdong-Hong Kong-Macao Applied Mathematics Center Project (Grant No. 2026A1515060008), and the Natural Science Foundation of Guangdong Province (Grant No. 2026A1515010171). The authors declare no competing interests.

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A Pre-Registered Hypotheses

Table 6 records all 16 pre-registered predictions alongside experimental outcomes. Predictions were added in five stages before running the corresponding experiments. The table is a selected scientific-claim ledger, not a consecutive experiment index: the first six rows retain the original unnumbered presentation, later rows retain historical experiment identifiers, and P7–P10 and P20 are not separately enumerated here. Numbering gaps do not represent additional outcomes counted in the total of 16.

Hosting and criteria. The ledger is hosted in the project notebook RESEARCH_JOURNEY_DEFENSE.md with stage-stamped entries; each stage was committed before the next round of training or evaluation began. A prediction is marked *confirmed* (✓) when its directional claim *and* any pre-stated quantitative threshold (e.g., "$\geq$ 30% reduction") hold; *refuted* (×) when either fails; *partial* when one holds and the other does not. Only secondary protocol details—mask seed indices, reporting-horizon choice, and figure formatting—were finalized post-hoc; the predictions themselves, including quantitative thresholds and the model bank, were frozen at registration time. Using the main-text categorization, the breakdown is ten confirmed, four refuted, and two partial; rows that miss the principal directional claim are counted as refuted even when a secondary threshold is informative. Verbatim re-statements are linked from each row of Table 6.

B Protocol and Transparency

B.1 Estimand-level crosswalk

Table 8 maps each reported estimand to its source table, units, normalization convention, and whether it is comparable across model families. Clean MAE is reported in canonical (mph) form for the primary table and in each family's standard masked convention for the extended bank; clean values are not directly cross-family-rankable because per-family training conventions differ. Shift estimands (Δ_{sensor}, cross-city, Δ_{temp}) are normalized so that within-family and (where matched structural access holds) across-family comparisons are sound.

The "Cross-family?" column reflects whether the estimand can be ranked across model families without re-introducing a convention artifact: clean MAE cannot, the sensor and city comparisons follow their stated normalization protocols, and the temporal ranking depends on the scaling and target-mask convention (Table 24). Routing concentration is compared across four row-stochastic backbones and extended to six models in Table 31.

B.2 Reproducibility, assets, and compute

All experiments use public traffic datasets and frozen checkpoints. The main numeric source for Tables 2–3 is `results/official_adaptive_relation/UPDATED_MERGED_COMPARISON_TABLE_2026-04-16.csv`; PEMS04 routing replication and mask-aware sweeps are read from `results/polish_v15/`. The checkpoint registry is `baselines/stgnn_protocol_registry.json`, and the unified training configuration is `configs/stgnn_unified.yaml`. Evaluation commands are listed in the repository README and mirror the commands used to generate the tables and figures. The code, configs, and frozen-checkpoint manifest are at `https://github.com/sirjuststop/stgnn-shift-bench`.

The released artifact contains the recommendation table and the scripts needed to regenerate the shift profiles from the canonical result files. It does not introduce a new dataset. METR-LA, PEMS-BAY, and PEMS04 remain under their original distribution terms; we cite the source papers and do not repackage the raw datasets in the submission. The runs were executed with one CUDA GPU per training/evaluation job. The April 16 extended-model evaluation log records roughly 3–9 minutes per seed for shift evaluation after training, depending on backbone; full retraining is not required to verify the frozen-checkpoint results.

C Sensor-Dropout Analysis

C.1 Severity curve: dropout ratios 10–50%

Table 9 reports Δ_{sensor} for three backbones across dropout ratios from 10% to 50%.

C.2 Additional corruption types

Table 11 reports the additional 30% corruption comparison on healthy sensors, alongside zero-fill. The evaluated types are stale/frozen hold, Gaussian noise, bias offset, and intermittent missingness. GWNet is sensitive to zero-fill and bias, but has the smallest noise increment among the models shown. Thus the silent-zero ranking does not extend unchanged to other corruptions. The reporting split matters too: under zero-fill, DCRNN's increase is $+0.39$ mph on healthy sensors and $+4.32$ mph on faulty sensors; GWNet's corresponding values are $+1.65$ and $+2.01$ mph. This comparison covers one severity (30%) and the healthy-sensor split. For zero-fill we also give the faulty-sensor values for DCRNN and GWNet; whole-network results and other severities for the non-zero corruptions are not tabulated.

C.3 Multi-mask aggregation

Table 12 reports Δ_{sensor} averaged over 5 random 30% zero-masks (seeds 42–46) per model checkpoint. The main-text results use the pre-registered primary mask per model seed. For DCRNN, MTGNN-Light, STGCN, and GWNet the 5-mask aggregate remains close to Table 2; GRU is a mask-sampling caveat.

C.4 Structured dropout patterns

Table 13 reports Δ_{sensor} under two topology-aware dropout patterns alongside the uniform random baseline. *Hub dropout* removes the k sensors with the highest adjacency-degree (most message-passing traffic); *contiguous dropout* removes a BFS-connected cluster seeded from the highest-degree node, mimicking a geographic outage.

The relative ordering GWNet $>$ STGCN $>$ DCRNN is preserved under all patterns. GWNet's hub-dropout sensitivity is **44% higher** than uniform random ($+0.127$ vs. $+0.088$, matched single-mask protocol shared with Table 2): hub nodes feed GWNet's densest adaptive-routing pathways, so removing them is especially damaging. DCRNN and STGCN show *lower* Δ_{sensor} under hub/contiguous dropout ($+0.008$, $+0.026$) than under uniform random dropout. When hubs are dropped, the kept sensors are peripheral nodes, which plausibly makes prediction easier. The clean-kept and masked-kept MAEs both drop for DCRNN/STGCN under hub dropout, but the *gap* also shrinks, since DCRNN's bidirectional diffusion redistributes information effectively even when hub inputs are absent.

C.5 Node-degree stratification

Table 14 reports Δ_{sensor} stratified by node degree quartile (Q1 = lowest degree, Q4 = highest). Nodes in each quartile are defined by their adjacency row-sum degree in the METR-LA graph. Only kept sensors (not dropped) are included in the MAE computation. Reported values are mean over 3 model seeds.

GWNet is the only backbone with a strongly monotonic degree→fragility pattern. MTGNNLight's weak positive relationship ($1.25\times$) and STGCN's intermediate ratio ($1.50\times$) show that some degree sensitivity exists in other models, but neither reaches GWNet's dense-adaptive-graph level ($1.83\times$), and only GWNet's increase is monotonic across Q1→Q4.

C.6 Mask-aware recovery: FP/FN sensitivity

The mask-aware adjacency variant zeroes out adjacency rows/columns for detected failed sensors. In silent-failure scenarios, the detector is imperfect: false positives (healthy sensors flagged as failed) and false negatives (failed sensors missed) can both occur. Table 15 reports MAE under the perfect mask vs. two realistic detector operating points.

Imperfect masks increase MAE relative to perfect detection: the mean penalty is $+0.044$ at FP5%/FN10% and $+0.077$ at FP10%/FN20%. The latter has substantial seed variation ($+0.049$–$+0.134$). We report these paired penalties directly; converting them to recovery percentages using the multi-mask baseline in Table 5 would mix protocols.

C.7 Mask-aware dropout-ratio sweep

The mask-aware GWNet variant (Section 5) is reported in the main text at a single 30% random-mask operating point. To test stability across dropout ratios, we sweep five ratios and add one structured pattern, using the true failure mask throughout. All cells use 3 GWNet seeds $\times$ 5 mask seeds; structured pattern uses the top-30% nodes by adjacency degree (one mask per seed).

The mask-aware recovery is in the 70–76% band across the 10–40% sweep (the 5% row's nominal 109% reflects an absolute $\Delta_{\text{sensor}}^{\text{mask-aware}}$ within seed noise of zero). Hub-dropout, where 30% of *high-degree* nodes fail simultaneously, is recovered at 71%—comparable to the random-30% main-text result—so the mitigation is not specific to the random mask used in the main text.

C.8 LOCF and mean imputation baselines

Table 17 compares three imputation strategies for dropped sensors: *zero-mask* (set dropped sensor channels to 0), *LOCF* (last observed value, i.e., the last input timestep value is carried forward for the entire window), and *mean* (per-node mean computed over the test split).

LOCF essentially eliminates the zero-mask degradation for all four backbones. Mean imputation performs similarly to or worse than zero-masking, because the test-split mean is influenced by the training distribution and creates a biased proxy in shifted traffic conditions. The LOCF result shows that the fragility ranking (GWNet $>$ STGCN $>$ DCRNN) is specific to zero-masking. Because the

comparison uses the true failure mask, it shows what carry-forward achieves when failed readings are identified, not what it achieves with an imperfect detector.

D Cross-City Transfer Analysis

D.1 Normalization conventions

Table 18 reports cross-city evaluation under both normalization conventions. *Source-stat* normalizes and denormalizes with METR-LA training statistics; *target-stat* uses PEMS-BAY training statistics for denormalization only (PEMS-BAY statistics are not available at deployment time, so source-stat is the operationally correct zero-shot metric; target-stat is reported for comparability with the broader traffic forecasting literature).

The source-stat improvement (PEMS-BAY $<$ METR-LA in normalized space) for all backbones reflects lower traffic variance on PEMS-BAY highway sensors, not better generalization. Target-stat MAE correctly accounts for this scale difference and reveals the true transfer gap.

D.2 Subset-size sensitivity

To test whether the GWNet transfer gap depends on how many PEMS-BAY sensors are used, we evaluate with 150-node coverage: 150 PEMS-BAY sensors are selected and the remaining 57 input slots are zero-padded (same protocol as sensor dropout). Table 19 reports city transfer MAE under both coverages.

The backbone ranking DCRNN $<$ STGCN $<$ GWNet is preserved at both coverage levels. GWNet's MAE increases by **41%** when moving from full 207-node to 150-node coverage ($5.30 \rightarrow 7.49$ mph); DCRNN decreases marginally (-2%) and STGCN decreases marginally (-3%). The large GWNet penalty arises because its source-city node embeddings route information through 57 zero-padded positions: inactive input slots inject systematic zero signals into the adaptive adjacency pathways, compounding the city-transfer gap. DCRNN and STGCN with target-city subgraph adjacency are unaffected by coverage reduction because zero-padded nodes are disconnected in the graph. Note: absolute MAE values differ from Table 2 due to different random subset draws; rankings are consistent.

D.3 Source-only vs. target adjacency

The main Table 2 uses a *matched structural constraint* (source-only adjacency) as the canonical cross-city comparison, so that no backbone benefits from access to target-city topology that others lack. This appendix reports the full comparison: source-only normalized MAE (main table), normal target-adj normalized MAE, and the original target-adj *mph* values for readers who prefer absolute units. GWNet and MTGNNLight are source-tied by construction (they learn or compute their own adjacency from source node embeddings); GRU uses no adjacency.

The source-only column (main table) tells the cleaner mechanistic story: DCRNN has lower MAE under the source-only constraint (0.180 vs. GWNet's 0.268, which is 49% higher), while STGCN's advantage over GWNet largely disappears (0.249 vs. 0.268). STGCN's city-transfer advantage in the target-adj column (0.202) arises from access to the target-city adjacency matrix, not from architectural generalization. DCRNN's bidirectional diffusion generalizes better even when confined to source topology, supporting its designation as the most transfer-robust backbone in the canonical bank; among all eight models, MegaCRN has the lowest source-stat MAE (Table 3).

D.4 Limited target-data adaptation

We relax the frozen rule for four canonical-bank backbones in a few-shot probe, fine-tuning the METR-LA checkpoints on $k \in \{50, 200, 800\}$ target windows for 30 steps. The 207-node PEMS-BAY subset is held fixed and inputs use target normalization. Table 21 includes the $k = 0$ starting point. The ordering does not change: DCRNN remains ahead, while GWNet improves only modestly. This supports the limited-adaptation comparison shown, rather than a claim about unrestricted fine-tuning. Its fixed-subset $k = 0$ values differ from Table 18, which averages five subsets; target-normalized input is also distinct from the source-normalized main-table setting.

D.5 PEMS04-source second city pair (P18)

To test whether DCRNN's transfer advantage is METR-LA-specific, we train the four variants shown here (DCRNN, GWNet, GWNet[Strong], GRU) on PEMS04 [Guo et al., 2019] (307 sensors, California District 4 freeway, 16,992 timesteps) with the same frozen training protocol and evaluate zero-shot on METR-LA and PEMS-BAY, using 5 random 307-node target subsets (seeds 42–46). The source PEMS04 adjacency is used for all backbones at inference, matching the P16 source-only fairness control. Additional STGCN and MTGNN checkpoints are included in the padding check below; the four rows here do not exhaust the PEMS04-source bank.

DCRNN has the lowest MAE in both PEMS04-source directions; on PEMS04→METR-LA, STGCN is close behind (7.18 vs. 7.01 mph, Table 23). Together with METR-LA→PEMS-BAY, the DCRNN advantage within these comparisons holds over three transfer directions among three cities, making a METR-LA-specific explanation unlikely. GWNet[Strong] is worse than default GWNet on PEMS04→METR-LA (12.58 vs. 11.39 mph), so extra adaptive-embedding capacity does not help here and may make transfer worse.

D.6 Cross-city zero-padding sensitivity

The cross-city eval in `experiments/stgnn_shift_eval_v5_pems04.py` (lines 87–91) uses index-repetition padding when target $N < N_{\text{source}}$: for PEMS04 (307) → METR-LA (207) the `keep` list is the 207 unique target nodes followed by 100 copies of node index 0, and MAE is averaged over all 307 prediction slots. The duplicate slots feed a single repeated input, which can either help (smoothing) or hurt (artificial concentration) the reported MAE. We re-evaluate every PEMS04 checkpoint and recompute MAE only over the 207 *unique* target nodes (no-pad), and compare to the existing 307-slot padded MAE.

The padding bias is small for the fixed-graph backbones DCRNN (-0.15 mph) and STGCN (-0.12 mph) but larger for GWNet, MTGNN, and GRU (-0.66, -0.73, -0.73 mph): index-repetition padding flatters the adaptive and graph-free models on PEMS04 → METR-LA by roughly 6–7% of their already-large MAE. The *ranking* is unchanged: DCRNN remains best in both columns, STGCN remains second (≈ 7 mph), and the other models stay above 10 mph. The PEMS04 → PEMS-BAY direction is unaffected since target $>$ source (subsample only, no padding).

E Temporal Shift Analysis

E.1 Absolute-unit weekday/weekend comparison

Table 24 pairs normalized and masked-mph results for the same three frozen checkpoints per model and the same test windows. MegaCRN and STAEformer change sign: their weekend error is lower in masked mph. Normalized MAE includes all targets, while the physical-unit metric excludes zero-valued targets. The comparison therefore changes both scaling and target masking; the observed reversal cannot be attributed solely to per-node standardization.

E.2 Per-horizon decomposition (P19)

Table 25 reports normalized MAE at horizons h=1, h=6, and h=12 under clean, 30% sensor-dropout, and cross-city (source-stat) evaluation (3 seeds, mean).

Under clean inference and sensor dropout, all backbones show the expected near-monotone MAE growth with horizon. Under cross-city transfer the models differ: DCRNN and STGCN retain normal growth (+46% and +33% from h=1 to h=12), whereas GWNet, GWNet[Strong], MTGNNLight, and GRU are nearly flat (+2%–+7%). For the adaptive-graph models, this pattern is consistent with a static routing mismatch rather than error that accumulates over the horizon; the horizon profile alone does not establish the cause.

E.3 Temporal covariate ablation

To test whether the normalized weekday/weekend gap of MegaCRN and STAEformer reflects an over-reliance on their frozen time-of-day (TOD) and day-of-week (DOW) embeddings, we run

two inference-time interventions across three seeds per model: (i) *randomize*: replace TOD/DOW channels with uniform noise; (ii) *swap*: swap covariates between weekday and weekend samples.

STAEformer shows partial covariate dependence; MegaCRN's gap is unchanged, so its normalized gap is not attributable to the frozen covariate channel alone. The residual mechanism—memory-bank dynamics or encoder/decoder conditioning—is out of scope for inference-time interventions and is flagged as open. These interventions concern the normalized metric and do not establish a higher weekend error in masked mph (Table 24).

F Routing Concentration: Mechanism and Replication

F.1 Adaptive adjacency density sweep

Table 27 reports the anchor-point density sweep ($k \in \{10, 20, \text{full}\}$) alongside GWNet[fixed] and MTGNNLight references. Table 28 extends this to the full k-grid $k \in \{5, 10, 15, 20, 30, \text{full}{=}207\}$.

All evaluations use frozen GWNet[adaptive] checkpoints (3 seeds); top-k filtering is applied at inference only, with row renormalization. Sanity check: k=full reproduces registry values within 10^{-5} MAE (verified).

The relative fragility trend is monotonically increasing across all six k values (5/5 consecutive pairs), supporting the association between denser adaptive connectivity and higher dropout fragility within this checkpoint. The absolute Δ_{sensor} values at k=5–20 are similar (0.072–0.074), with the gap widening at k=30–full. As noted in Section 5.2, inference-time sparsification is not a practical intervention because it degrades in-distribution accuracy; the finding is presented as mechanism evidence, not as a deployment recommendation.

F.2 Routing-sparsity spectrum across adaptive models

Table 29 reports the per-model adjacency concentration and sensor-dropout delta for the four original adaptive-graph models. The proxies average three METR-LA-trained checkpoints. max_mass is the mean over rows and samples of the largest entry in each adjacency row; eff_k is $\exp(\text{row-entropy})$ and measures effective support size.

STAEformer is excluded from the Pearson because its spatial mixing is head-stratified multi-head attention rather than a single row-stochastic adjacency: a max_mass extracted from any one head is not commensurable with the per-row probability used in the other four backbones, so the proxy-isomorphism assumption underlying the correlation does not hold. The four-backbone Pearson is therefore reported on the row-stochastic subset where the proxy is identically defined.

The routing-spectrum deltas differ from the primary-mask values in Tables 2–3 (AGCRN $+0.019$, MegaCRN -0.048, GWNet $+0.088$). Using those primary-mask deltas with the same saved concentration means gives Pearson $+0.880$, so the association's strength depends on the delta version. MegaCRN's -0.040 in Table 34 comes from a separate ablation evaluation and is not substituted into either correlation.

Under the conservative ranking that treats MegaCRN's negative delta as an adjacency-smoothing artifact (Section 5.3), max_mass and Δ_{sensor} are monotonic across all four models.

F.2.1 Seed-level variation and rank-correlation diagnostics

Per-seed max_mass values for the four adaptive backbones are AGCRN $\{0.0077, 0.0076, 0.0100\}$ (population std 0.0011), MegaCRN $\{0.0064, 0.0093, 0.0097\}$ (0.0015), MTGNN-Light $\{0.1586, 0.1607, 0.1596\}$ (0.0009), and GWNet $\{0.1638, 0.2773, 0.2475\}$ (0.0480). All four spreads are below the MTGNN-Light–GWNet mean gap (0.070). With GTS (0.288) and DGCRN (0.325), the upper-end gaps are 0.058 and 0.037; the latter is smaller than GWNet's population std, so the order of the most concentrated models may vary across seeds.

Per-seed Pearson against Δ_{sensor} is $\{+0.92, +0.99, +0.98\}$ across the three seeds (mean $+0.96$, population std 0.03 from these rounded values), suggesting that the max_mass relation is not an artifact of seed averaging despite GWNet's larger within-seed spread. Spearman over the four seed-mean values is $\rho_S{=}+0.80$; the single rank exchange swaps AGCRN (0.0085) and MegaCRN

(0.0085), whose max_mass values differ by less than 10^{-3}, well below either backbone's per-seed std. Treating the two memory-meta backbones as tied on the proxy yields rank correlation $\rho_S{=}{+}1.0$ across the three distinguishable groups $\{$tied AGCRN/MegaCRN, MTGNN-Light, GWNet$\}$.

F.2.2 Leave-one-out and exact permutation analysis

Using the unrounded saved max_mass values and the routing-spectrum deltas in Table 29, the four-model Pearson is $+0.978827$. Table 30 gives all leave-one-out results. Removing GWNet leaves $+0.960752$, so the positive association does not disappear when that model is excluded. Enumerating all $4!{=}24$ permutations of the four deltas, and counting $|\rho_{\mathrm{perm}}| \geq |\rho_{\mathrm{obs}}|$, gives $p{=}2/24{=}0.0833$. The architecture-level sample remains four models.

F.3 Six-architecture extension and native-source summaries

GTS [Shang et al., 2021] and DGCRN [Li et al., 2023] extend the four-backbone comparison with link-predictor and input-conditioned graph learning. Their METR-LA max_mass values are 0.288 and 0.325, above GWNet's 0.230. The six-architecture Pearson is $+0.974$, $+0.973$ without GWNet, and minimum leave-one-out $+0.961$. Table 31 reports approximate two-sided Pearson t tests calculated as $t = r\sqrt{(n-2)/(1-r^2)}$ from the rounded coefficients and $n = 6$ (4 degrees of freedom). Neither PEMS-BAY nor PEMS04 reaches $p < 0.05$ under this test. These six-model summaries differ from the original four-model PEMS04 replication below; the two sample sizes should not be conflated.

F.4 PEMS04-native routing-concentration replication

To address whether the routing-concentration relation generalizes beyond a single source city, we re-train the four adaptive-graph backbones on PEMS04 (307 sensors, chronological 70/10/20 split) using the same training configuration as the canonical METR-LA bank (Section 3.2). max_mass is then recomputed from each PEMS04-trained checkpoint via the same per-architecture adjacency proxy (Section 5.2), and Δ_{sensor} is measured on PEMS04 test data (3 model seeds $\times$ 5 mask seeds, 30% zero-mask, MAE on kept sensors only).

Two findings support the shared-mechanism hypothesis. (i) Both source cities separate the low max_mass of AGCRN/MegaCRN from the higher values of MTGNN/GWNet. (ii) MegaCRN's METR-LA-specific negative Δ_{sensor} (-0.005) becomes positive on PEMS04 ($+0.057$), consistent with the adjacency-mediated weekday-prior smoothing diagnosis in Section 5.3. This is consistent with the routing-concentration account, although four models on one additional dataset cannot establish a shared mechanism.

F.5 Theory sketch: first-order exposure bound

Setup. Let a single message-passing layer produce node representations $h_i = \sum_j A_{ij} W x_j$ where $A \in \mathbb{R}^{N \times N}$ is row-stochastic and W is a linear map. Perturbing the inputs at nodes in $\mathcal{I} \subseteq [N]$ by zero-masking, the change in h_i is $\Delta h_i = -\sum_{j \in \mathcal{I}} A_{ij} W x_j$.

Exposure bound. Taking norms and writing $\|x\|_\infty := \max_j \|x_j\|_2$ for the largest node-feature norm:

$$\|\Delta h_i\|_2 \leq \|W\|_2 \cdot \|x\|_\infty \cdot \sum_{j \in \mathcal{I}} A_{ij}.$$

For any row i, let $m_i = \max_j A_{ij}$. Then $\sum_{j \in \mathcal{I}} A_{ij} \leq |\mathcal{I}| m_i$, so

$$\|\Delta h_i\|_2 \leq \|W\|_2 \cdot \|x\|_\infty \cdot |\mathcal{I}| m_i.$$

Averaging over target nodes yields

$$\frac{1}{N} \sum_i \|\Delta h_i\|_2 \leq \|W\|_2 \cdot \|x\|_\infty \cdot |\mathcal{I}| \cdot \text{max_mass}(A).$$

Interpretation. The bound says that concentrated rows have larger worst-case exposure when the dropped set includes high-mass neighbors. It is deliberately local: it does not prove full-horizon MAE behavior and does not cover nonlinear recurrence, temporal convolutions, or decoder amplification. It also should not be read as an exact expected-error result under iid dropout. If each sensor is dropped independently with probability p, then for a row-stochastic A,

$$\mathbb{E}_{\mathcal{I}}\left[\sum_{j\in\mathcal{I}} A_{ij}\right] = p\sum_{j} A_{ij} = p,$$

which is independent of m_i. Thus max_mass(A) is best viewed as a tail-risk or worst-case exposure proxy. The bound is also loose at the dropout rates used here: for $|\mathcal{I}| \approx 62$ (30% of 207 sensors), $|\mathcal{I}|m_i > 1$ whenever $m_i > 0.016$, whereas the mean row maxima for MTGNN-Light, GWNet, GTS and DGCRN are roughly 10–20 times larger. For such rows it is weaker than the trivial bound $\sum_{j\in\mathcal{I}} A_{ij} \leq 1$. The argument motivates measuring max mass; it does not account for the size of the correlations in Tables 29, 31, and 32. The mask-aware intervention (Section 5) targets the same exposure term by zeroing and renormalizing routes attached to known dropped sensors.

G Model Ablations

G.1 GWNet capacity variants (P17 and P21)

We train GWNet[Strong] with hidden dim 128, adaptive embed dim 20, and early-stopping patience 30 (otherwise identical to the standard config). Table 33 compares it to the default GWNet and DCRNN across all shift axes. We additionally report GWNet[Best] (P21), the best configuration identified by in-distribution MAE across a 12-config Stage-1 grid (hidden $\in \{128, 256\}$, embed $\in \{20, 32\}$, lr $\in$ {1e-3, 5e-4, 2e-4}, dropout $\in \{0.1, 0.3\}$); the winning setting is hidden 256, embed 32, lr 5e-4, patience 30.

Doubling hidden dimension and adaptive embedding closes 34% of the in-distribution MAE gap to DCRNN ($7.85 \rightarrow 7.42$ mph) and halves the sensor-dropout Δ ($+0.088 \rightarrow +0.050$), matching the sparse MTGNNLight baseline. However, the cross-city MAE is effectively unchanged ($4.42 \rightarrow 4.46$ mph), the source-only adjacency gap to DCRNN persists (0.262 vs 0.180), and the degree-stratified hub-fragility signature remains monotonic ($1.72\times$ Q4/Q1, still far from DCRNN's $0.95\times$).

The full 12-config sweep GWNet[Best] (hidden 256, embed 32, lr 5e-4, patience 30) further reduces the in-distribution gap (7.22 ± 0.04 mph, closing about half of the gap to DCRNN) and the sensor-dropout fragility ($+0.042 \pm 0.002$), yet Δ_{sensor} remains $2.1\times$ DCRNN's $+0.020$—confirming P21. The source-only city-transfer normalized MAE (0.262 ± 0.009) is identical to GWNet[Strong]'s (0.262) and still 46% above DCRNN's 0.180. Capacity attenuates but does not eliminate the architectural fragility, supporting the mechanism interpretation in Section 5.

G.2 MegaCRN memory-bank ablation

To test whether MegaCRN's meta-memory bank absorbs missing sensors as a form of implicit regularization, we zero-mask and re-route its attention/memory paths at inference in four modes. *full*: the unmodified model; *memzero*: query the memory bank with a zero vector so the prototype fall-back is fixed regardless of input; *graphonly*: zero out the attention output keeping the adjacency pathway; *attnonly*: use identity adjacency while keeping memory attention. We report Δ_{sensor} means over three seeds and five dropout ratios $r \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$.

memzero deepens rather than flattens the U-shape, refuting the memory-bank regularization hypothesis. **attnonly** (identity adjacency) eliminates Δ_{sensor} completely, indicating that cross-node flow through the adjacency is the dominant driver in this ablation. Per-seed σ at r=0.2 is 0.031, approximately equal to the mean—the apparent robustness is seed-fragile. The mechanism is adjacency-mediated smoothing: zero-diffused predictions drift toward a weekday-mean prior that accidentally tracks the test set.

H Architecture-Selection Decision Tree

Table 35 reports the conservative $2\times2\times2$ recommendation table over the deployment profile. The tree is generated by `experiments/decision_tree_artifact.py` from the same per-model normalized fragility vectors used in the main paper. The conservative ranking demotes flagged *(model, axis)* pairs where low apparent fragility is an artifact (*(megacrn, sensor)*; see Section 5.3). The temporal axis uses the original normalized gaps. Its weekend-related recommendations are conditional on that metric; physical-error comparisons are given separately in Table 24.

Under these normalized profiles, four of the eight cells reach DCRNN, reflecting its uniformly low fragility across axes. MegaCRN is recommended only under the single-shift city-transfer profile; STAEformer wins the sensor-only cell; AGCRN wins the two weekend-exposed cells without city transfer. The tree and supporting scripts are included in the repository artifact (`stgnn-shift-bench`) so that practitioners can reproduce the recommendation and the underlying rankings on new frozen checkpoints.

Table 6: Pre-registered hypotheses and outcomes. Confirmed (✓), partially confirmed, or refuted (×) by experiment.

Prediction	**Outcome**	**Result**
GWNet fails hardest under city transfer (topology overfitting)	GWNet is worst under sensor dropout, not city transfer; GRU is worst at city transfer	×
Graph-based models are more robust than GRU to sensor dropout	GRU is nearly as robust as DCRNN at 30%; adaptive graph hurts, fixed graph helps	Partial ×
All models degrade on PEMS-BAY vs. METR-LA	Source-stat MAE improves on PEMS-BAY (simpler traffic); target-stat reveals real gaps	×
Adaptive adjacency drives sensor-dropout fragility ($\geq$30% reduction when ablated)	43.1% reduction confirmed	✓
Temporal shift magnitude differs across architectures	All architectures within one std of each other	×
STGCN (fixed spectral adj) $\Delta_{\text{sensor}} \leq$ DCRNN (fixed diffusion adj)	STGCN $+0.078 >$ DCRNN $+0.020$; but both $<$ GWNet $+0.088$. Adaptive adj is the dominant driver.	×
P11–P13 added prior to structured / coverage experiments:		
[P11] GWNet transfer gap persists (GWNet worst) across 150- and 207-node PEMS-BAY coverage	Confirmed: ranking preserved; GWNet +41% MAE at 150 nodes vs. -2% for DCRNN	✓
[P12] Hub dropout raises GWNet Δ_{sensor} relative to uniform random (dense routing)	Confirmed: GWNet $+44\%$ ($+0.127$ vs. $+0.088$); DCRNN lower ($+0.008$ vs. random $+0.020$)	✓
[P13] Mask-aware recovery meaningful with FP $\leq 5\%$, FN $\leq 10\%$ detection	Partial: error increases under imperfect detection; high variance at FP=10%/FN=20%	Partial ✓
P14–P16 added prior to inductive-baseline and capacity-matched experiments:		
[P14] MTGNNLight (sparse cosine-sim) shows lower Δ_{sensor} than GWNet (dense product)	Confirmed: MTGNNLight $+0.051$, GWNet $+0.088$ (42% lower)	✓
[P15] GWNet[CM] (doubled patience) shows same Δ_{sensor} as standard GWNet	Confirmed: $+0.088$ vs $+0.088$ (identical); fragility is architectural not training artifact	✓
[P16] Source-only adj control narrows but does not eliminate DCRNN's transfer advantage	Confirmed: DCRNN 0.180 vs GWNet 0.268 (49% gap remains under fair comparison)	✓
P17–P19 added prior to stronger-GWNet, second-city-pair, and per-horizon experiments:		
[P17] GWNet[Strong] (hidden 128, embed 20) still shows $\Delta_{\text{sensor}}>2\times$ DCRNN and unchanged city MAE	Confirmed: $+0.050$ vs $+0.020$ ($2.5\times$); city 4.46 vs 4.42 mph	✓
[P18] DCRNN is best-generalizing on ≥ 1 PEMS04-source direction (PEMS04$\rightarrow$METR-LA, PEMS04$\rightarrow$PEMS-BAY)	Confirmed on *both*: DCRNN 7.01/2.65 mph; STGCN is close on METR-LA (7.18; Table 23)	✓
[P19] GWNet's per-horizon city-transfer MAE is flatter than DCRNN's (static mis-routing, not accumulation)	Confirmed: GWNet $+4\%$ h1$\rightarrow$h12 growth vs DCRNN $+46\%$	✓
P21 added prior to full-sweep best-config experiment:		
[P21] GWNet[Best] (12-config sweep: hidden 256, embed 32, lr 5e-4) still shows $\Delta_{\text{sensor}}>2\times$ DCRNN	Confirmed: $+0.042$ vs $+0.020$ ($2.1\times$); in-dist 7.22 mph; tuning over a 12-configuration grid does not remove the fragility	✓

Table 7: Key supplementary checks behind the main findings. Each row points to the corresponding appendix experiment.

Experiment	Key result	Ref.
GWNet[Best] 12-config sweep (P21)	$\Delta_{\text{sensor}} = +0.042$ (2.1× DCRNN); the gap persists after tuning	G.1
PEMS04 cross-city (P18)	DCRNN lowest in both directions: 7.01/2.65 mph; STGCN is close on METR-LA (7.18 mph)	D.5
Source-adj vs target-adj	STGCN's apparent advantage shrinks (0.249 src-only vs 0.202 target-adj)	D.3
Hub dropout	GWNet $+44\%$ ($+0.127$ vs. matched $+0.088$); DCRNN -60% ($+0.008$ vs. $+0.020$)	C.4
Per-horizon city MAE (P19)	GWNet: $+4\%$ h1→h12 growth vs DCRNN $+46\%$	E.2

PEMS04 cross-city values are denormalized mph; all Δ values are normalized.

Table 8: Estimand-level crosswalk: which numbers are cross-family-comparable.

Estimand	Source	Unit	Normalization	Cross-family?
Clean MAE	T.2, T.3	mph (canon.) / masked (ext.)	per-family	No
Δ_{sensor}	T.2, T.3	normalized	source-stat	Yes (within-family)
Cross-city MAE	T.2, T.3	normalized	source / target (both)	Yes (matched-access)
Δ_{temp}	T.2, T.3, T.24	normalized / masked mph	per-model	Metric-dependent
Routing concentration	T.29, T.31	unitless $[0, 1]$	per-row	Proxy-matched; n=4, extended n=6

Table 9: **GWNet is most fragile in this three-model comparison; the gap grows nonlinearly.** Δ_{sensor} mean over 3 seeds across dropout ratios 10–50%. GWNet/DCRNN ratio (bottom) quantifies relative fragility.

Backbone	10%	20%	30%	40%	50%
DCRNN	$+0.005$	$+0.012$	$+0.020$	$+0.033$	$+0.044$
GWNet	$+0.013$	$+0.044$	$+0.088$	$+0.149$	$+0.189$
GRU	$+0.001$	$+0.005$	$+0.025$	$+0.073$	$+0.088$
GWNet/DCRNN	$2.5\times$	$3.8\times$	$4.3\times$	$4.6\times$	$4.3\times$

Table 10: Physical-unit zero-fill severity comparison, healthy-sensor split. GTS and DGCRN extend the concentration range; routing-mass values are rounded. DCRNN uses a fixed graph.

Backbone	Max mass	10%	20%	30%	40%	50%
DCRNN	—	$+0.07$	$+0.21$	$+0.39$	$+0.60$	$+0.81$
MTGNN-Light	0.16	$+0.19$	$+0.54$	$+0.99$	$+1.54$	$+2.22$
GWNet	0.23	$+0.32$	$+0.96$	$+1.65$	$+2.67$	$+3.47$
GTS	0.29	$+0.41$	$+0.96$	$+1.75$	$+2.53$	$+3.26$
DGCRN	0.32	$+0.38$	$+1.51$	$+2.85$	$+4.92$	$+6.34$

Table 11: Additional sensor corruptions on the canonical bank and GWNet[Strong] control: 30% affected channels, healthy-sensor split. Entries are MAE increases in mph relative to clean inputs.

Backbone	Zero	Stale	Noise	Bias	Intermittent
DCRNN	$+0.39$	$+0.18$	$+0.28$	$+1.00$	$+0.21$
GRU	$+0.88$	$+0.58$	$+0.20$	$+1.02$	$+0.25$
STGCN	$+1.50$	$+0.32$	$+1.34$	$+4.24$	$+0.80$
MTGNN-Light	$+0.99$	$+0.30$	$+0.29$	$+2.12$	$+0.52$
GWNet	$+1.65$	$+0.28$	$+0.16$	$+1.96$	$+0.73$
GWNet[Strong]	$+1.49$	$+0.39$	$+0.29$	$+3.12$	$+0.78$

Table 12: Multi-mask sensor dropout: Δ_{sensor} mean and std over 5 random masks (seeds 42–46), then averaged over 3 model seeds. Most values are consistent with primary-mask results in Table 2; GRU is the exception noted in the protocol.

Backbone	Δ_{sensor} (multi-mask mean)	Seed std	vs. Table 2
DCRNN (fixed diffusion)	$+0.021$	± 0.002	$+0.020$ (consistent)
GRU (graph-free)	$+0.044$	± 0.002	$+0.025$ (seed-42 was low-variance mask)
MTGNNLight (sparse cosine-sim)	$+0.051$	± 0.012	$+0.051$ (consistent)
STGCN (fixed spectral)	$+0.079$	± 0.006	$+0.078$ (consistent)
GWNet (adaptive adj)	$+0.092$	± 0.030	$+0.088$ (consistent; high mask variance)
GWNet[CM] (patience=30)	$+0.088$	± 0.017	$+0.088$ (P15 confirmed identical)

Table 13: Structured vs. uniform sensor dropout (Δ_{sensor}, mean over 3 seeds, 30% drop rate). Hub: top-k degree sensors dropped. Contiguous: BFS cluster from highest-degree seed. GWNet ordering is preserved (>STGCN>DCRNN); structured patterns amplify GWNet's fragility but reduce DCRNN/STGCN's apparent sensitivity, because hub dropout keeps peripheral sensors that may be easier to predict.

Backbone	Random (uniform)	Hub dropout	Contiguous dropout
DCRNN (fixed diffusion)	$+0.020$	$+0.008$	$+0.005$
STGCN (fixed spectral)	$+0.078$	$+0.026$	$+0.022$
GWNet (adaptive adj)	$+0.088$	$+0.127$	$+0.096$

Table 14: Sensor-dropout Δ_{sensor} by node-degree quartile (30% zero-mask, kept sensors only, 3-seed mean). Mean node degrees: Q1 $\bar{d} = 1.83$, Q2 $\bar{d} = 3.32$, Q3 $\bar{d} = 4.48$, Q4 $\bar{d} = 5.98$.

Backbone	Q1 (low)	Q2	Q3	Q4 (high)	Q4/Q1
DCRNN	$+0.019$	$+0.026$	$+0.018$	$+0.018$	$0.95\times$
MTGNNLight	$+0.044$	$+0.052$	$+0.050$	$+0.055$	$1.25\times$
STGCN	$+0.059$	$+0.089$	$+0.074$	$+0.089$	$1.50\times$
GWNet	$+0.066$	$+0.076$	$+0.085$	$+0.121$	$1.83\times$

Table 15: Mask-aware GWNet sensitivity to imperfect detection (3 model seeds). FP is the fraction of healthy sensors erroneously zeroed in the adjacency; FN is the fraction of failed sensors missed. Values are the normalized-MAE increase relative to perfect detection under the same primary mask, reported as mean $\pm$ population std and median across seeds.

Detector scenario	FP / FN	Mean $\pm$ std	Median
Perfect mask	0%/0%	$+0.000 \pm 0.000$	$+0.000$
Conservative detector	5%/10%	$+0.044 \pm 0.017$	$+0.045$
Noisy detector	10%/20%	$+0.077 \pm 0.040$	$+0.049$

Table 16: Mask-aware GWNet recovery is stable across dropout ratios and structured patterns. "Standard" = silent zero-mask; "mask-aware" = adjacency rows/cols zeroed and renormalized for dropped sensors; recovery $= 1 - \Delta^{\text{mask-aware}}_{\text{sensor}} / \Delta^{\text{standard}}_{\text{sensor}}$. Script: `experiments/mask_aware_dropout_sweep.py`; data: `results/polish_v15/mask_aware_ratio_sweep.json`.

Cell	$\Delta^{\text{std.}}_{\text{sensor}}$	$\Delta^{\text{mask-aware}}_{\text{sensor}}$	Recovery
Random 5%	$+0.004 \pm 0.002$	$+0.000 \pm 0.001$	$109\% \pm 29$
Random 10%	$+0.019 \pm 0.006$	$+0.005 \pm 0.002$	$76\% \pm 6$
Random 20%	$+0.051 \pm 0.013$	$+0.015 \pm 0.002$	$70\% \pm 5$
Random 30% (main)	$+0.092 \pm 0.030$	$+0.023 \pm 0.009$	$75\% \pm 3$
Random 40%	$+0.145 \pm 0.026$	$+0.035 \pm 0.006$	$75\% \pm 3$
Hub-30% (structured)	$+0.116 \pm 0.037$	$+0.034 \pm 0.001$	71%

Hub selection differs from Table 13: this sweep ranks by the count of nonzero neighbors, whereas that table ranks by the sum of adjacency weights. The resulting failure sets differ, accounting for the distinct standard deltas (0.116 vs. 0.127).

Table 17: Sensor-dropout Δ under three imputation strategies (30% drop rate, seed=42, 3 model seeds). LOCF improvement: fraction of zero-mask gap closed by LOCF. Negative improvement indicates worse than zero-mask.

Backbone	Δ_{zero}	Δ_{LOCF}	Δ_{mean}	**LOCF improvement**
DCRNN	$+0.020$	≈ 0	$+0.025$	$\sim 102\%$
MTGNNLight	$+0.051$	≈ 0	$+0.077$	$\sim 100\%$
STGCN	$+0.078$	$+0.003$	$+0.079$	$\sim 97\%$
GWNet	$+0.088$	≈ 0	$+0.109$	$\sim 101\%$

Table 18: Cross-city MAE under two normalization conventions, using target adjacency for DCRNN/STGCN (not the source-only main comparison). Source-stat: METR-LA train mean/std for both normalization and denormalization (lower = PEMS-BAY traffic is statistically simpler in normalized space). Target-stat: PEMS-BAY train statistics for denormalization (interpretable mph units). Mean $\pm$ population std over 5 random 207-node subsets and 3 model seeds.

Backbone	**Clean MAE (METR-LA, norm)**	**Source-stat MAE (norm)**	**Target-stat MAE (mph)**
DCRNN (fixed diffusion)	0.359	0.168	2.74 ± 0.01
STGCN (fixed spectral)	0.388	0.209	3.27 ± 0.02
GWNet (adaptive adj)	0.427	0.268	4.42 ± 0.15
GRU (graph-free)	0.408	0.278	4.73 ± 0.19

Table 19: Cross-city transfer MAE (target-stat, mph) under variable sensor coverage. 150-node: 150 PEMS-BAY sensors active, 57 input slots zero-padded to 207. 207-node: full model capacity (different subset draws from main table; see text). Mean $\pm$ std over 3 random PEMS-BAY subsets, 3 model seeds.

Backbone	**150-node coverage**	**207-node (full)**
DCRNN (fixed diffusion)	4.25 ± 0.03	4.33 ± 0.05
STGCN (fixed spectral)	4.94 ± 0.25	5.07 ± 0.33
GWNet (adaptive adj)	7.49 ± 1.08	5.30 ± 0.25

Table 20: Full cross-city transfer comparison. *Source-only (norm)*: all models constrained to METR-LA adjacency; canonical main-table values. *Target-adj (norm)*: DCRNN/STGCN use PEMS-BAY adjacency (GWNet/MTGNNLight/GRU unchanged). *Target-adj (mph)*: same target-adj evaluation, denormalized using PEMS-BAY statistics. Mean $\pm$ population std over 5 PEMS-BAY subsets, 3 model seeds.

Backbone	**Source-only (norm) [main table]**	**Target-adj (norm)**	**Target-adj (mph)**
DCRNN (fixed diffusion)	0.180 ± 0.001	0.167 ± 0.002	2.74 ± 0.01
MTGNNLight (cosine-sim)	0.244 ± 0.004	0.244 ± 0.004	4.15 ± 0.11
STGCN (fixed spectral)	0.249 ± 0.008	0.202 ± 0.005	3.27 ± 0.02
GRU (graph-free)	0.278 ± 0.009	0.278 ± 0.009	4.73 ± 0.19
GWNet (dense adaptive)	0.268 ± 0.006	0.268 ± 0.006	4.42 ± 0.15

Table 21: Limited target-data adaptation: target-test MAE in mph on one fixed 207-node PEMS-BAY subset, using target normalization. This is separate from the five-subset source-normalized main comparison.

Model	$k = 0$	$k = 50$	$k = 200$	$k = 800$
DCRNN	2.69	2.69	2.66	2.67
MTGNN-Light	4.13	3.82	3.79	3.78
GWNet	4.22	4.06	4.13	4.11
GRU	4.44	4.25	4.21	4.18

Table 22: PEMS04-source cross-city transfer (3 seeds $\times$ 5 target subsets, mean $\pm$ std). PEMS04 in-dist MAE in native flow units; target MAE in target-stat mph. DCRNN has the lowest MAE in both directions; STGCN is close on METR-LA when included in Table 23.

Backbone	PEMS04 in-dist	$\rightarrow$ METR-LA (mph)	$\rightarrow$ PEMS-BAY (mph)
DCRNN	26.19 ± 0.02	$\mathbf{7.01 \pm 0.09}$	$\mathbf{2.65 \pm 0.02}$
GWNet	33.47 ± 0.45	11.39 ± 0.62	4.58 ± 0.16
GWNet[Strong]	32.62 ± 0.17	12.58 ± 0.32	4.61 ± 0.02
GRU	25.81 ± 0.06	11.03 ± 0.01	4.68 ± 0.03

Table 23: Cross-city MAE (mph) on METR-LA target with the existing index-repetition padding vs. a no-pad variant that scores only the 207 unique target nodes. PEMS04 $\rightarrow$ PEMS-BAY (N=325) is unaffected since target $>$ source (subsample only). 3 PEMS04 seeds $\times$ 5 target subsets. Script: `experiments/cross_city_no_pad_sensitivity.py`; data: `results/polish_v15/cross_city_no_pad_sensitivity.json`.

Backbone	Padded MAE (mph)	No-pad MAE (mph)	Δ (pad $-$ nopad)
DCRNN	7.014 ± 0.085	7.166 ± 0.088	-0.152
STGCN	7.176 ± 0.169	7.295 ± 0.158	-0.118
GWNet	11.390 ± 0.623	12.051 ± 0.422	-0.661
MTGNN	10.292 ± 0.090	11.020 ± 0.083	-0.728
GRU	11.026 ± 0.008	11.756 ± 0.012	-0.730

Table 24: Weekday/weekend gap under two reporting conventions. The same checkpoints and test windows are used in both columns; normalized MAE uses all targets, while masked mph excludes zero targets. Negative gaps mean lower weekend error. Mean $\pm$ population std over three seeds.

Model	Normalized/plain gap	Masked-mph gap
AGCRN	$+0.061 \pm 0.008$	$+0.750 \pm 0.013$
MegaCRN	$+0.734 \pm 0.019$	-0.594 ± 0.051
STAEformer	$+0.750 \pm 0.003$	-0.612 ± 0.002

Table 25: Per-horizon normalized MAE on METR-LA at h=1, 6, 12. Clean and 30% sensor dropout computed on the METR-LA test set; city-src computed on the 5-subset PEMS-BAY evaluation. Δh: relative growth from h=1 to h=12.

	Clean			Masked 30%			City-src		
Backbone	h_1	h_6	h_{12}	h_1	h_6	h_{12}	h_1	h_6	h_{12}
DCRNN	0.251	0.356	0.448	0.356	0.449	0.527	0.133	0.160	0.194
MTGNNLight	0.291	0.370	0.450	0.371	0.444	0.515	0.237	0.240	0.253
STGCN	0.296	0.385	0.466	0.438	0.513	0.578	0.173	0.199	0.230
GRU	0.331	0.403	0.474	0.375	0.435	0.499	0.263	0.269	0.282
GWNet	0.363	0.423	0.485	0.464	0.516	0.568	0.261	0.264	0.272
GWNet[Strong]	0.333	0.401	0.468	0.390	0.457	0.522	0.254	0.257	0.260

Table 26: Weekday $\rightarrow$ weekend Δ_{temp} (normalized) under covariate interventions. Script: `experiments/temporal_covariate_ablation.py`.

Model	Clean	Randomize	Swap	Reduction (swap)
MegaCRN	0.7336	0.7254	0.7336	0.0%
STAEformer	0.7495	0.7212	0.6556	12.5%

Table 27: Density sweep anchor points: relative fragility increases monotonically with adaptive connectivity density. Rel. fragility $= \Delta_{\text{sensor}}/\text{MAE}_{\text{in-dist}}$. Mean $\pm$ std over 3 seeds; Δ_{sensor} averaged over 5 random 30% masks.

Variant	In-dist MAE	Δ_{sensor} (30%)	Rel. fragility
GWNet[fixed]	8.27 ± 0.08	$+0.050 \pm 0.010$	0.60%
GWNet[k=10]	9.03 ± 0.42	$+0.073 \pm 0.030$	0.80%
GWNet[k=20]	8.85 ± 0.34	$+0.074 \pm 0.032$	0.84%
GWNet[k=full]	7.85 ± 0.11	$+0.092 \pm 0.030$	1.17%
MTGNNLight (ref)	6.88 ± 0.11	$+0.051 \pm 0.012$	0.74%

Table 28: Full k-grid density sweep for GWNet[adaptive] on METR-LA. Mean $\pm$ std over 3 seeds. Δ_{sensor}: averaged over 5 random 30% zero-masks (seeds 42–46). Rel. fragility $= \Delta_{\text{sensor}}/\text{MAE}_{\text{in-dist}}$. Note: k<full disrupts embeddings trained for dense routing, causing in-distribution degradation at low k; the clean-performance cost at k=5 is $+17\%$ relative to k=full.

k	In-dist MAE	Δ_{sensor} (30%)	Rel. fragility
5	9.21 ± 0.50	$+0.072 \pm 0.031$	0.78%
10	9.03 ± 0.42	$+0.073 \pm 0.030$	0.80%
15	8.94 ± 0.38	$+0.073 \pm 0.031$	0.81%
20	8.85 ± 0.34	$+0.074 \pm 0.032$	0.84%
30	8.66 ± 0.30	$+0.078 \pm 0.033$	0.89%
full $= 207$	7.85 ± 0.11	$+0.092 \pm 0.030$	1.17%

Table 29: Routing concentration and sensor-dropout sensitivity across four adaptive models. Pearson is $+0.979$ with the routing-spectrum deltas and $+0.880$ with the primary-mask deltas. Script: `experiments/routing_sparsity_spectrum.py`.

Model	Family	max_mass	eff_k	density	Δ_{sensor}
AGCRN	dense-symmetric	0.008	≈ 200	1.00	$+0.011$
MegaCRN	memory-meta	0.008	≈ 200	1.00	-0.005
MTGNN-Light	sparse-top-k	0.160	—	sparse	$+0.051$
GWNet	dense-bilinear	0.230	≈ 114	1.00	$+0.092$

Table 30: Leave-one-out Pearson correlations for the original four-model routing-spectrum estimates. The full-sample Pearson is $+0.978827$; the exact absolute-value-tail permutation test over all 24 permutations gives $p = 0.0833$.

Excluded model	Remaining-model Pearson
AGCRN	$+0.992784$
MegaCRN	$+0.976678$
MTGNN-Light	$+0.988090$
GWNet	$+0.960752$

Table 31: Six-architecture native-source routing-concentration correlations. Approximate two-sided Pearson t tests (p_t, 4 degrees of freedom) use rounded correlations. Test details: Appendix F.3.

Source dataset	Quantity	n	Pearson	Approx. p_t
METR-LA	Speed	6	$+0.974$	0.001
PEMS-BAY	Speed	6	$+0.750$	0.086
PEMS04	Flow	6	$+0.630$	0.180

Table 32: Routing-concentration relation replicates on PEMS04. Pearson $\rho(\text{max_mass}, \Delta_{\text{sensor}}) = +0.910$ across the four adaptive backbones (METR-LA reference: $+0.979$). Native checkpoints; same protocol as Table 29. Script: `experiments/pems04_routing_gate_native.py`; data: `results/polish_v15/pems04_native_routing_gate.json`.

	METR-LA (reference)		PEMS04 (native)	
Model	**max_mass**	Δ_{sensor}	**max_mass**	Δ_{sensor}
AGCRN	0.008	$+0.011$	0.006 ± 0.000	$+0.051 \pm 0.006$
MegaCRN	0.008	-0.005	0.005 ± 0.000	$+0.057 \pm 0.019$
MTGNN	0.160	$+0.051$	0.145 ± 0.001	$+0.119 \pm 0.004$
GWNet	0.230	$+0.092$	0.248 ± 0.020	$+0.114 \pm 0.026$

Table 33: GWNet capacity scaling: standard, P17 strong, and P21 best-sweep configs vs. DCRNN (3 seeds, mean $\pm$ std). Source-only adjacency MAE follows the P16 fairness control protocol. Q4/Q1 is the degree-stratified fragility ratio (Appendix C.5). GWNet[Best] city MAE (mph) and Q4/Q1 were not separately evaluated in the sweep; source-only normalized values are reported.

Variant	**In-dist (mph)**	Δ_{sensor} **(30%)**	**City MAE (mph)**	**Source-only adj (norm)**	**Q4/Q1 ratio**
DCRNN	6.58 ± 0.03	$+0.020 \pm 0.002$	2.74 ± 0.01	0.180 ± 0.001	$0.95\times$
GWNet	7.85 ± 0.11	$+0.088 \pm 0.017$	4.42 ± 0.15	0.268 ± 0.006	$1.83\times$
GWNet[Strong]	7.42 ± 0.04	$+0.050 \pm 0.018$	4.46 ± 0.06	0.262 ± 0.004	$1.72\times$
GWNet[Best]	7.22 ± 0.04	$+0.042 \pm 0.002$	—	0.262 ± 0.009	—

Table 34: MegaCRN Δ_{sensor} (normalized) by dropout ratio and ablation mode. Script: `experiments/megacrn_memory_ablation.py`.

Mode	r=0.1	r=0.2	r=0.3	r=0.4	r=0.5
full	-0.001	-0.043	-0.040	-0.016	$+0.045$
memzero	-0.006	-0.067	-0.075	-0.038	-0.007
graphonly	$+0.001$	-0.027	-0.024	$+0.012$	$+0.068$
attnonly	0.000	0.000	0.000	0.000	0.000

Table 35: Architecture recommendation per deployment profile. Margin is the difference in weighted-risk score to the runner-up (lower = closer call). Confidence is margin normalized by runner-up risk. AGCRN, recommended in the two weekend-exposed profiles without city transfer, has the only positive masked-mph weekend gap among the extended models (Table 24); these cells should not guide selection for physical-error performance.

sensor	**city**	**weekend**	**Recommended**	**Runner-up**	**Confidence**
no	no	no	DCRNN	AGCRN	0.30
no	no	yes	AGCRN	MTGNN-Light	1.00
no	yes	no	MegaCRN	DCRNN	1.00
no	yes	yes	DCRNN	AGCRN	0.51
yes	no	no	STAEformer	AGCRN	0.18
yes	no	yes	AGCRN	DCRNN	0.19
yes	yes	no	DCRNN	MegaCRN	0.34
yes	yes	yes	DCRNN	AGCRN	0.30

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   Answer: [Yes].

   Justification: The abstract and introduction state that the paper is a pre-registered diagnostic of frozen traffic STGNNs, not a new model or dataset. The scope is limited to the evaluated shifts, source cities, and masking assumptions.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes].

   Justification: Section 6 covers highway loop-detector data, the three-seed main bank, zero-shot transfer as a diagnostic floor, within-split temporal comparisons, silent zero-masking, and the detection requirement for mitigation. Appendix B.2 describes the released assets.

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